Noa Moriel — Curriculum Vitæ

Contact Information

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Professional Summary

Research scientist with 7 years in machine learning across industry and academia (Riverside, IBM Research, Mobileye, Harvard). Works on foundation models, agentic pipelines and learning the dynamics of complex systems. Computer Science Ph.D. at the Hebrew University. Publications at ICLR, Nature Biotechnology and Nature Protocols.

Experience

Riverside, *NLP Research Engineer* 2025–Present
Built agentic and ML pipelines that automatically edit recorded episodes.

IBM Research, *Research Scientist* 2024–2025
Quantified confidence in transcriptomic foundation models (NeurIPS Workshop 2025).

Harvard University, *Research Scientist* 2019–2020
Reconstructed tissue structure from single-cell data with optimal transport (NovoSpaRc).

Mobileye, *Algorithm Engineer* 2018–2019
Built real-time deep learning detection and scene-understanding models for autonomous vehicles.

Broad Institute of MIT and Harvard, *Research Scientist* 2016
Adapted single-cell RNA-seq and its analysis to yeast; designed a combinatorial perturbation library.

Education

Hebrew University, Jerusalem, Israel.
Computer Science and Computational Biology.
Ph.D. Advised by Mor Nitzan. 2020–2025
M.Sc. Advised by Nir Friedman. *2nd in class; GPA 95.1/100.* 2016–2018
B.Sc. *Magna Cum Laude; GPA 94.2/100.* 2012–2015
Council of Higher Education and CIDR Ph.D. Fellowships; Dean's Award.
M.Sc. exchange at University of Toronto.
Teaching assistant: Machine Learning, AI, Genomics, Algorithms in Computational Biology.

Selected Papers

† First or co-first author. Full list on Google Scholar.

- Characterizing Nonlinear Dynamics via Smooth Prototype Equivalences — PRX Life 2026
- Toward a Coherent Virtual Cell Model† — NeurIPS Workshop 2025
- Time-Warp-Attend: Learning Topological Invariants of Dynamical Systems† — ICLR 2024
- Optimal Sequencing Budget Allocation for Trajectory Reconstruction† — ISMB 2024
- TACCO: Annotation Transfer and Decomposition of Cell Identities† — Nature Biotech. 2023
- Emergence of Division of Labor in Tissues through Cell Interactions† — Cell Reports 2023
- Phase2vec: Dynamical Systems Embedding with a Physics-Informed CNN — ICLR 2023
- NovoSpaRc: Spatial Reconstruction of Single-Cell Gene Expression† — Nature Protocols 2021

Military Service

Pilot training; Unit 8200, Israel Defense Forces. Intelligence Excellence Award. 2010–2012